

RAMBØLL	PROJECT —	PROJECT NO. —	PREPARED —
	Crack Width Verification - RC Slab (EN 1992-1-1 7.3.4)		CHECKED —
REV / DATE 0.1 · 2026-07-20			APPROVED —
			DRAFT

PURPOSE

Characteristic crack width for a reinforced concrete slab under quasi-permanent loading.

1 Input Parameters

Symbol	Value	Unit	Description
b	1000	mm	Section width (per metre strip)
h	300	mm	Section depth
d	260	mm	Effective depth
c_{nom}	30	mm	Nominal concrete cover
ϕ	16	mm	Bar diameter
A_s	1005	mm ²	Tension reinforcement area (5 H16)
M_{qp}	45	kNm	Quasi-permanent bending moment
E_s	200000	MPa	Steel modulus of elasticity
E_{cm}	33000	MPa	Concrete secant modulus
$f_{ct, eff}$	2.9	MPa	Effective concrete tensile strength
k_t	0.4	-	Load duration factor (long term)
k_1	0.8	-	Bond factor (high-bond bars)
k_2	0.5	-	Strain distribution factor (bending)
k_3	3.4	-	Code coefficient
k_4	0.425	-	Code coefficient
w_{max}	0.3	mm	Allowable crack width (XC exposure)

2 Calculation

Modular ratio E_s/E_{cm}

$$\alpha_e = \frac{E_s}{E_{cm}} = \frac{200000}{33000} = 6.061$$

Neutral axis depth (cracked section)

$$x = \frac{A_s \cdot \alpha_e \cdot (\sqrt{\frac{2 \cdot b \cdot d}{A_s \cdot \alpha_e} + 1} - 1)}{b} = \frac{1005 \cdot 6.061 \cdot (\sqrt{\frac{2 \cdot 1000 \cdot 260}{1005 \cdot 6.061} + 1} - 1)}{1000} = 50.52 \text{ mm}$$

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Internal lever arm $z = d - \frac{x}{3}$ $= 260 - \frac{50.52}{3}$ $= \mathbf{243.2 \text{ mm}}$
Steel stress under quasi-permanent loads $\sigma_s = M_{qp} \cdot 1000000 \cdot \frac{1}{A_s \cdot z}$ $= 45 \cdot 1000000 \cdot \frac{1}{1005 \cdot 243.2}$ $= \mathbf{184.1 \text{ MPa}}$
Effective tension zone height $h_{c, \text{eff}} = \min(\frac{h}{2}, \frac{h-x}{3}, 2.5 \cdot (-d + h))$ $= \min(\frac{300}{2}, \frac{300-50.52}{3}, 2.5 \cdot (-260 + 300))$ $= \mathbf{83.16 \text{ mm}}$
Effective reinforcement ratio $\rho_{p, \text{eff}} = \frac{A_s}{b \cdot h_{c, \text{eff}}}$ $= \frac{1005}{1000 \cdot 83.16}$ $= \mathbf{0.01208}$
Maximum crack spacing $S_{r, \text{max}} = C_{nom} \cdot k_3 + \frac{k_1 \cdot k_2 \cdot k_4 \cdot \phi}{\rho_{p, \text{eff}}}$ $= 3.4 \cdot 30 + \frac{0.425 \cdot 0.5 \cdot 0.8 \cdot 16}{0.01208}$ $= \mathbf{327.1 \text{ mm}}$
Mean strain difference (eps_sm - eps_cm) $\epsilon_{diff} = \max(\frac{0.6 \cdot \sigma_s}{E_s}, \frac{-\frac{f_{ct, \text{eff}} \cdot k_t \cdot (\alpha_e \cdot \rho_{p, \text{eff}} + 1)}{\rho_{p, \text{eff}}}}{E_s} + \sigma_s)$ $= \max(\frac{0.6 \cdot 184.1}{200000}, \frac{184.1 - \frac{0.4 \cdot 2.9 \cdot (0.01208 \cdot 6.061 + 1)}{0.01208}}{200000})$ $= \mathbf{0.0005524}$
Characteristic crack width $w_k = \epsilon_{diff} \cdot S_{r, \text{max}}$ $= 0.0005524 \cdot 327.1$ $= \mathbf{0.1807 \text{ mm}}$

3 Verification

OK	$w_k = 0.1807 \text{ mm} \leq w_{max} = 0.3 \text{ mm}$
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